

# GuardShield AI: Viva / Jury Examination Q&A; & Research Trends

## Comprehensive Examiner Question Preparation Guide & Academic Literature Evolution

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### Q1: Why build a custom proxy instead of using Meta's Llama-Guard-3?

**Answer:** Llama-Guard is a full 8B LLM adding 1-2s latency. GuardShield fine-tunes DeBERTa-v3 (86M) adding sub-25ms overhead.

### Q2: How do you handle prompt injections in Hinglish or Base64?

**Answer:** DeBERTa-v3 operates on subword BPE tokenization and contextual embeddings, recognizing intent even when obfuscated.

### Q3: How does GuardShield detect hallucinations in real-time?

**Answer:** We calculate Shannon Entropy over streaming token probabilities. If entropy spikes, we trigger a windowed NLI check mid-sentence under 25ms.

### Q4: Did you buy any hardware?

**Answer:** No! 100% software-based requiring 0 cost. Everything runs locally on standard laptop GPUs via Ollama and PyTorch.